

Microsoft Fabric Practical Summary

Objective: To design and implement a scalable, automated data pipeline and analytics architecture using Microsoft Fabric.

This practical bridges raw multi-source data extraction with enterprise-grade dimensional modeling, demonstrating a complete Medallion Architecture lifecycle from ingestion to governance.

1. Infrastructure Provisioning & Unified Storage

- **Fabric Setup & Foundation:**
 - Architected the foundational environment by configuring the Fabric workspace and navigating tenant and capacity concepts.
 - Established a robust workspace-level organization strategy while leveraging **OneLake's** decoupled storage behavior to eliminate redundant data movement.
- **Multi-Source Integration:**
 - Successfully integrated disparate data ecosystems, practicing ingestion from external GitHub-based API files and Azure Data Lake Storage **(ADLS) Gen2**.
 - Secured all cross-platform data access by configuring Role-Based Access Control **(RBAC)** and enterprise connectors.

2. Automated Data Ingestion Layer

- **Orchestration via Data Factory:**
 - Built and orchestrated **Fabric Data Factory** pipelines to seamlessly extract diverse file formats from both **web-based API sources** and legacy Azure Data Lake storage, landing the raw data directly into the Fabric Lakehouse.
- **Dynamic Pipeline Design:**
 - Bypassed brittle, hardcoded ETL methods by engineering a metadata-driven ingestion architecture.
 - Utilized a master **JSON configuration** file combined with **Lookup** and **ForEach** activities to dynamically iterate through and process multiple files autonomously.

3. Data Processing & The Medallion Architecture

- **Lakehouse Engineering:**
 - Implemented core **Medallion Architecture** principles (Bronze and Silver layers) within the Fabric Lakehouse.
 - Raw source files were systematically stored in the Bronze layer, while cleaned and transformed datasets were written into processed Delta tables in the Silver layer.
- **Cloud-Native Transformations:**
 - Bridged intuitive data wrangling with cloud compute using **Data Flow Gen2**.
 - Executed **low-code** transformations (including cross-table joins, row filtering, duplicate handling, and conditional column generation) before loading the optimized output back into the Lakehouse.
- **Code-First PySpark Engineering:**
 - Leveraged **Apache Spark** for heavy-duty big data processing.
 - Developed **PySpark** notebooks to read raw CSV and Delta files, dynamically infer schemas.
 - Created derived columns, split complex fields, and union fragmented yearly sales data before writing the final results into **ACID-compliant Delta Parquet tables**.

4. Enterprise Data Warehousing & Modeling

- **Lake-Centric Warehousing:**
 - Designed a **Fabric Data Warehouse** specifically optimized for business intelligence consumption.
 - Defined a dedicated **Gold schema** and utilized **CTAS** (Create Table As Select) operations to rapidly instantiate warehouse tables directly from the Lakehouse managed tables without duplicate data loading.
- **Dimensional Modeling:**
 - Transitioned processed data into an optimized Star Schema format to support high-performance analytical queries.
 - Modelled and populated a central fact_sales table surrounded by conformed dimension tables, including dim_customers, dim_products, and dim_calendar.

5. CI/CD & Data Governance

- **Lineage & Auditing:**
 - Explored and utilized **Fabric's visual lineage** view to rigorously track data flow dependencies across Data Factory pipelines, Lakehouses, PySpark notebooks, SQL Warehouses, and final **Semantic Models**.
- **Deployment & Lifecycle Management:**
 - Practiced continuous integration and deployment principles using Fabric Deployment Pipelines, successfully managing the safe promotion of data artifacts across isolated Development, Testing, and Production environments.